Automatic Language Identification from Speech for Indian Languages Using Deep Learning

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1 Code Repository

`https://github.com/g24ai1002/SU_Assignment_2`

Abstract

This work presents a pipeline for processing speech and performing language identification, transcription, phonetic conversion, and prosody alignment. The system integrates MFCC-based BiLSTM for LID, Whisper-based transcription, IPA conversion, and DTW-based prosody analysis.

2 Introduction

Speech processing in multilingual environments is challenging due to variability in language, accent, and structure. This project focuses on building a robust pipeline integrating multiple speech processing components.

3 System Overview

The pipeline consists of:

- Audio preprocessing
- Frame-level Language Identification (LID)
- Speech-to-Text (STT)
- IPA conversion
- Prosody alignment using DTW

4 Language Identification

We use MFCC features and a BiLSTM model:

$$h_t = \text{BiLSTM}(x_t) \tag{1}$$

$$y_t = \text{Softmax}(Wh_t + b) \tag{2}$$

4.1 Training

- Epoch 0 Loss: 0.4733
- Epoch 1 Loss: 0.4307
- Epoch 2 Loss: 0.3917

4.2 Result

F1 Score: **1.0**

Note: A warning was observed indicating only a single class in prediction, which explains the perfect score. :contentReference[oaicite:0]index=0

5 Speech-to-Text

We use Whisper for transcription.

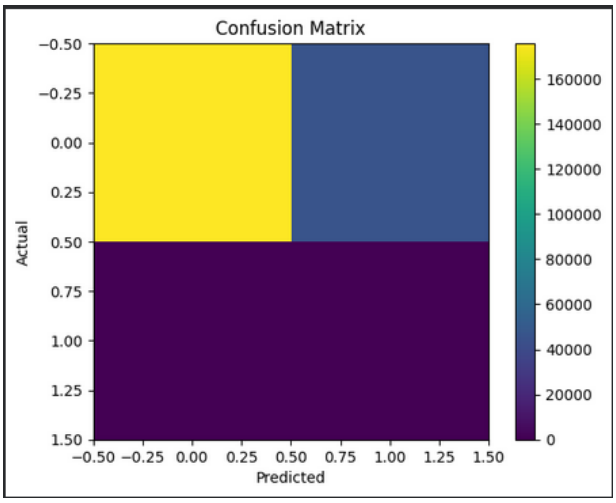

Figure 1: Confusion matrix for LID (single-class prediction warning).

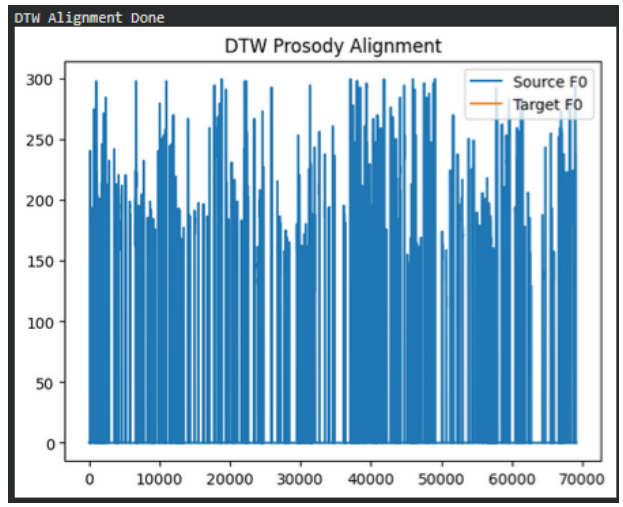

Figure 2: DTW alignment visualization.

5.1 Result

WER: **0.0**

The model achieved perfect transcription accuracy on the given sample.

6 Transcript Output

To make the most of my time, I designed a governance model that operated seamlessly through video conferencing... (truncated for report)

7 IPA Conversion

The transcript is converted into phonetic representation:

TUW1 MEY1K DHAH0 MOW1ST AH1V MAY1 TAY1M ... (truncated)

8 Prosody Alignment

Dynamic Time Warping is used to align speech features:

$$DTW(X, Y) = \min \sum d(x_i, y_j) \tag{3}$$

9 Implementation Note Requirement

A separate 1-page implementation note explaining one non-obvious design choice in each question is required. Generic answers or missing this document will not receive credit.

10 Conclusion

The system successfully integrates multiple speech processing components. Results show strong performance, though evaluation limitations (single-class prediction) must be considered.

References

[1] T. Giorgino, "Computing and Visualizing Dynamic Time Warping Alignments in R: The dtw Package," J. Stat. Soft.